

SManager

Multi-Server Dashboard System

INTRODUCTION



A lightweight distributed system for centralized server management,
monitoring based on remote task execution.

SYSTEM COMPONENTS



WEB GUI

Modern, responsive user interface for server orchestration and monitoring.



CORE

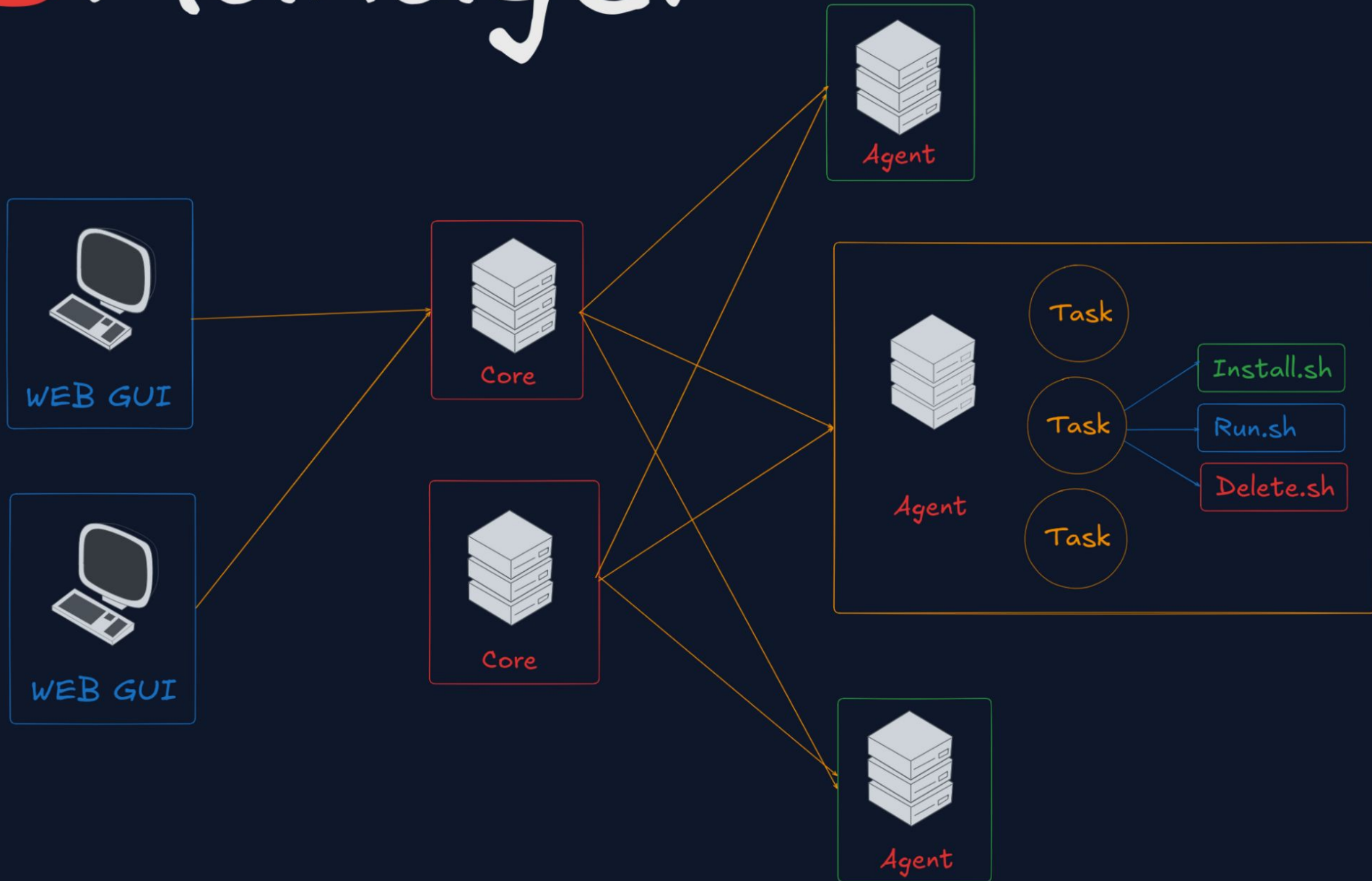
The central brain handling data distribution and agent synchronization.



AGENT

Lightweight service deployed on remote servers for task execution.

SManager



| RUST CONCEPTS: PERFORMANCE

-  **Async/Await + Tokio:** Leveraging non-blocking I/O operations to handle massive concurrency without overhead.
-  **Ownership & Borrowing:** Ensuring compile-time memory safety and preventing common bugs like data races.
-  **Smart Pointers:** Utilization of Arc for safe, atomic thread-shared data access.

| RUST CONCEPTS: PERFORMANCE

 Traits - Code reuse via Clone, Serialize, and Deserialize for efficient data handling.

 Pattern Matching - Robust and safe packet parsing using Rust's powerful match expressions.

 Error Handling - Explicit error management using Result types for highly stable services.

CORE LIBRARY ECOSYSTEM

Tokio - async

Serde - Json

Chrono - Time

Base64 - encoding hashes and tokens

Sqlx - databases

Dotenvy - Env files

Rustls-pemfile, rustls, X509-parser - mTLS connection for Agent-Core

Sha2 - hashes

Jsonwebtoken - for ui

log / env_logger - logging

Nix - Unix API

Tokio-tungstenite - websockets

| SECURITY & CRYPTO

Securing the Pipeline

The security of this application is based on multiple layers of security at each stage of communication.

 **mTLS:** Mutual TLS via Rustls & X509-parser for encrypting and authenticating agent-core communication.

 **JWT:** Token-based authentication for UI-Core communication.

 **Hashing:** Hashing for encoding sensitive data using salt.

 Validation clients by CN, ip and connection order.

 Constantly keeping heartbeat between each system element.

CORE CHALLENGES



Async Complexity

Orchestrating multiple tasks simultaneously while maintaining safety and performance in a high-concurrency environment.



mTLS Implementation

Establishing robust and secure mutual TLS connections between Agents and the Core for strict security measures.

Questions?

Thank you for your attention to the SManager.